

UCSD Machine Learning Bootcamp Capstone Research Survey Writeup

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Using a multilingual Large Language Model (MLLM), I created an app that tutors its users on the Swedish language for my capstone project. My research focused on learning the underlying algorithms behind an MLLM and creating a capstone prototype. To this end, I consulted the following papers:

1) "Lost in Translation: Large Language Models in Non-English Content Analysis" [1]

This paper provides background on the motivation behind Multilingual LLMs, a high-level overview behind how they differ from regular LLMs, and key challenges. It discusses the distinction between high-resource languages (eg. English, Spanish, Chinese, etc.) and low-resource languages (eg. Armenian, Creole, etc.) and why that distinction matters. Swedish is a medium-resource language with a well-established system of datasets (eg. Språkbanken) but lacking in the widespread LLM adoption of its high-level counterparts.

2) "Attention Is All You Need" [2]

This is the iconic paper that began the modern age of the LLMs. Its unique contribution to machine learning is that of the transformer architecture, which utilizes the self-attention algorithm to resolve blocks of text in parallel unlike its predecessors (recurrent neural networks).

3) "How do Large Language Models Handle Multilingualism?" [3]

This is another key paper discussing the similarities of the training processes between LLMs and MLLMs (which mainly uses a larger dataset consisting of multiple languages). It explores the concept of cross-attention, which is analogous to self-attention but between two different languages in an MLLM. In practice, MLLMs translate the input into English, handle all the reasoning in intermediary neural network layers, and then translate the output back into the original language. The authors also performed an experiment to identify language-specific neurons in different MLLMs. By turning off all but a few neurons at a time and measuring the MLLM's performance, the paper's authors identified which neurons of various MLLMs correlate to which languages.

4) "Multilingual Large Language Models: A Systematic Survey" [4]

This paper is a reference on the history of MLLMs thus far, the different types of MLLM architectures, the different corporas and crawlers used to train them, and their applications to biology, medicine, computer science, mathematics, and law.

This prototype leverages prompt engineering to wrap a tutoring user-interface around an existing MLLM. It questions the user in Swedish, accepts a response, and provides feedback in

English on the user's Swedish mistakes. The key challenges are to form suitable prompts, to display the MLLM's responses to the user, and to measure the MLLM's effectiveness as a Swedish tutor. My initial prototype succeeds in the first two goals but falls short on the third. The app sends two prompts: one to take a topic of the user's choice and ask a question and another to provide feedback to the user based on the question from the previous prompt and the user's response. I downloaded the MLLM from Ollama while my prototype interacts with its api to send and receive prompts. I implemented the user-interface interface with Python and Flask. I chose Llama 3.1 as the MLLM for the prototype.

The prototype attempts to measure Llama 3.1 as a Swedish tutor by asking it to answer test questions from a dataset series known as SuperLIM. [5] SuperLIM provides multiple test types ranging from "word in context" questions to multiple-choice analogies. The "word in context" question dataset gives the same Swedish word in two different sentences and asks if the word holds the same definition in both sentences. Running Llama 3.1 against the part of my code measuring its performance in the 1000 "word in context" questions yields a confusion matrix indicating an overall accuracy score between 55 and 60% (multiple runs yield different accuracies in that range). The problem with the Super-LIM tests is that they measure an MLLM's general comprehension of the Swedish language, but not necessarily its ability to tutor a user. Although Språkbanken contains dozens of question and answer datasets in Swedish, none include feedback statements addressed towards deliberately incorrectly written Swedish sentences (eg. users who forgot to add "på" in "Jag tittar på TV:n."). Therefore, I am required to create my own testing data. I will accomplish this in the next iteration in a future step of this capstone project. Strong test criteria provide the backbone for how one defines and solves larger data science projects, thereby making it more effective to start with them and later build the app rather than the other way around.

Works Referenced

[1] "Lost in Translation: Large Language Models in Non-English Content Analysis".
<https://arxiv.org/abs/2306.07377>

[2] ""Attention Is All You Need". <https://arxiv.org/abs/1706.03762>

[3] "How do Large Language Models Handle Multilingualism?". <https://arxiv.org/abs/2402.18815>

[4] "Multilingual Large Language Models: A Systematic Survey".
<https://arxiv.org/abs/2411.11072>

[5] "Super-LIM". <https://spraakbanken.gu.se/resurser/superlim>